REGULAR ARTICLE

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Hydroxyurea pharmacokinetics in children with sickle cell anemia across different global populations

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Key Points

- Interpatient differences in hydroxyurea dosing for children with SCA are predicted by drug PK.
- Hydroxyurea PK parameters in children with SCA from the United States, Africa, and the Caribbean predict similar maximum tolerated doses.

Hydroxyurea provides effective disease-modifying treatment for people with sickle cell anemia (SCA), especially when escalated to maximum tolerated dose (MTD), which has wide interpatient dosing variability due to pharmacokinetic (PK) differences. Whether hydroxyurea PK parameters differ among children with SCA in different global regions is unknown. We compared hydroxyurea PK parameters among children with SCA from 5 clinical trials: HUSTLE (United States), TREAT (United States), NOHARM (Uganda), REACH (Uganda and Kenya), and EXTEND (Jamaica). Key hydroxyurea PK parameters were determined using HdxSim, a validated hydroxyurea PK software program. The results were compared across regions by analysis of variance. PK profiles from 451 children with SCA (146 from the United States, 265 from Africa, and 40 from the Caribbean) were included. Children from Africa had slightly lower volumes of distribution, but absorption rate and clearance were similar across regions. The PK-recommended doses to achieve MTD were statistically different but clinically similar across the United States (26.6 ± 5.9 mg/kg per day), Africa (27.6 ± 6.5 mg/kg per day), and the Caribbean (25.2 ± 4.7 mg/kg per day) (P = .04). In multivariable regression, younger age and increased reticulocyte counts were associated with higher PK-recommended doses. Hydroxyurea PK parameters in children with SCA differ minimally across global populations, predicting clinically similar doses to achieve MTD. Individualized hydroxyurea dosing based on a PK-population model derived from US children with SCA can be used broadly to maximize the benefits of this critical medication in other global populations. These trials were registered at www.ClinicalTrials.gov as #NCT00305175 (HUSTLE), #NCT02286154 (TREAT), #NCT01976416 (NOHARM), #NCT01966731 (REACH), and #NCT02556099 (EXTEND).

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The data generated and analyzed during this study are included in published article. Further inquiries can be directed to the corresponding author, Alexandra Power-Hays (alexandra.power-hays@cchmc.org).

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Introduction

Hydroxyurea is a safe and effective oral medication for people living with sickle cell anemia (SCA) that has been a pivotal disease-modifying treatment for this vulnerable patient population and is now the standard-of-care in many countries worldwide.[1-5] Hydroxyurea provides multiple benefits by increasing protective fetal hemoglobin (HbF), improving red blood cell rheology, and reducing inflammation. There are various published hydroxyurea dosing strategies including fixed, weight-based dosing at low (~10 mg/kg per day) or moderate ranges (~20 mg/kg per day),[6-9] as well as dose escalation to achieve the maximum tolerated dose (MTD), which is the individualized daily dose that maximizes HbF induction and provides mild myelosuppression, but without laboratory or clinical toxicities.[5,10,11] Clinical trials in children with SCA have found that hydroxyurea treatment at MTD has superior clinical efficacy compared with fixed, weight-based dosing. Treatment at MTD leads to increased HbF and reduced anemia, hospitalizations, blood transfusions, and stroke risk via improvement in transcranial Doppler velocities.[12-16]

There is wide interpatient variability in hydroxyurea MTD,[1,17,18] and determining an individual's optimal dose can be logistically challenging to patients and providers due to the need for serial blood draws and dose adjustments.[19-23] The variation in hydroxyurea MTD among individuals with SCA is, in part, related to individual differences in the drug's absorption, distribution, and clearance.[17,24-26] These parameters are collectively referred to as pharmacokinetics (PK), and they affect the duration and concentration of hydroxyurea to which the bone marrow and other tissues are exposed. Prospective trials in children with SCA in the United States, using personalized dosing of hydroxyurea based on individual PK parameters and target hydroxyurea exposure, have resulted in decreased time to achieving MTD and higher HbF values without increased laboratory toxicity.[27,28] In many cases, the HbF levels in those started on their individualized dose exceeded 30% with a near-pancellular distribution, and were comparable to levels obtained in successful gene therapy trials.[29]

Over 75% of people with SCA live in sub-Saharan Africa and other low-resource settings, but hydroxyurea PK studies have been conducted primarily in small cohorts of people with SCA living in North America and Europe.[17,30-36] Different nutritional, environmental, genetic, and clinical factors are likely to affect hydroxyurea PK parameters. To expand our understanding of hydroxyurea dosing and considering the potential for improved dose determination using PK guidance, it is critical to compare hydroxyurea PK parameters in people with SCA living in different global regions. In this study, hydroxyurea PK profiles from 5 distinct SCA cohorts of children enrolled on clinical trials in the United States, Jamaica, Uganda, and Kenya were compared. Furthermore, laboratory and clinical variables were analyzed for their impact on the PK parameters and PK-recommended hydroxyurea doses.

Methods

Overview

We performed a cross-sectional comparison of the hydroxyurea PK profiles of children with SCA who had PK samples collected during prospective, open-label hydroxyurea clinical trials over the past 15 years. PK profiles from participants enrolled in HUSTLE (Memphis, United States), TREAT (Cincinnati, United States), NOHARM (Kampala, Uganda), REACH (Mbale, Uganda; and Kilifi, Kenya), and EXTEND (Kingston, Jamaica) trials were included (Table 1). All PK collections and analyses were approved by the respective institutional review board and ethics committees for each trial.

PK parameters and recommended dose

The details of PK collection in each trial are summarized in Table 1. After providing written informed consent, participants underwent

Table 1. Summary of the 5 clinical trials analyzed for hydroxyurea PK parameters

	HUSTLE	TREAT	NOHARM	REACH	EXTEND
NCT number	NCT00305175	NCT02286154	NCT01976416	NCT01966731	NCT02556099
Location	Memphis, United States	Cincinnati, United States	Kampala, Uganda	Mbale, Uganda Kilifi, Kenya	Kingston, Jamaica
Year published	2011	2019	2017	2019	2021
Age of eligibility, y	1-30	6 mo to 21 y	1.0-3.9	1-10	2-17
Age at PK collection, mean ± SD, y	8.7 ± 4.5	3.3 ± 5.0	3.8 ± 0.9	8.1 ± 2.3	9.7 ± 2.6
Dose timing	First dose or on treatment	First dose	On treatment	On treatment	On treatment
Dose administered, mg/kg	20	20	Current dose	Current dose	Current dose (MTD)
Dose administered, mean ± SD, mg/kg	20.2 ± 1.8	19.9 ± 0.5	26.5 ± 6.2	22.2 ± 4.7	24.3 ± 7.1
Blood collection	Venipuncture	Volumetric collection device and venipuncture	Volumetric collection device	Volumetric collection device	Volumetric collection device
Collection sample	Serum	Whole blood or plasma	Whole blood	Whole blood	Whole blood
Sparse sampling time points, min	30, 60, 240	20, 60, 180	15, 60, 180	15, 60, 180	15, 60, 180
Hydroxyurea analysis	Spectrophotometer	LC-MS	LC-MS	LC-MS	LC-MS

LC-MS, liquid chromatography–mass spectrometry; NCT, National Clinical Trial identifier.

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PK collection involving the administration of a single oral hydroxyurea dose followed by timed blood sample collection.[26] The hydroxyurea dose administered varied based on trial methods. Among the TREAT and HUSTLE "old cohort" participants, the dose administered before PK testing was their first dose. Among the NOHARM, REACH, and HUSTLE "new cohort" participants, the dose administered was the currently tolerated dose. In EXTEND, the dose administered was their clinical MTD based on dose escalation toward target myelosuppression.[37]

In the HUSTLE trial, samples were collected ≤8 hours after the dose.[17] A sparse sampling strategy was previously determined via the analysis of the HUSTLE results through optimal sampling theory, which concluded that only 3 time points were necessary for accurate hydroxyurea PK evaluation.[26] These time points include an early collection (15-30 minutes), middle (60 minutes), and a later time point (3-4 hours). For this analysis, sparse time points identified through the optimal sampling theory were used for the PK profiles from the HUSTLE participants; prior analysis has shown no difference in PK parameters between sparse and rich sampling from this cohort.[26,38] Subsequent trials used these sparse sampling strategies (Table 1). PK profiles from all participants with 3 timed hydroxyurea samples available for this analysis were included.

The hydroxyurea concentration in each timed sample was quantified using validated analytical methods.[18] The samples from HUSTLE were analyzed via spectrophotometry in Memphis, United States.[17] The remainder samples were analyzed via liquid chromatography–mass spectrometry in Cincinnati, United States.[18]

Hydroxyurea concentration-time profiles from participants were analyzed using a validated hydroxyurea PK software program (HdxSim; Medimatics, version 2.3.2.96).[38] HdxSim was designed with funding from the National Institutes of Health to increase access to PK-guided hydroxyurea dosing in places that do not have clinical pharmacology support. It is a user-friendly software program that uses a population PK model in combination with Bayesian estimation to determine individual PK parameters based on individual patient demographics and timed hydroxyurea concentration results.[26,38] The population PK model was designed from the rich sampling from the HUSTLE cohort that determined that pediatric hydroxyurea PK were best described by a 1-compartment model with transit absorption and Michaelis-Menten elimination.[26] The program estimates the area under the time-concentration curve (AUC) and the optimal dose to reach individual MTD. The target hydroxyurea exposure was set at an AUC of 115 mg•h/L, which was the mean AUC achieved by the HUSTLE cohort at clinical MTD.[17] This target AUC was subsequently shown to be safe and effective when prospectively validated for PK-guided dosing of hydroxyurea in the TREAT and HOPS trials (www.ClinicalTrials.gov identifier: NCT03789591), which were conducted in the United States.[27,28]

For this analysis, the AUC, PK-recommended dose, absorption rate, volume of distribution, and clearance rate were determined via HdxSim for each child. The ratio of AUC to test dose was calculated as AUC divided by test dose. These PK parameters were compared across regions: NOHARM and REACH comprises the Africa group, EXTEND comprises the Caribbean group, and HUSTLE and TREAT comprise the US group.

Evaluation of clinical variables

The associations between selected demographic and clinical variables with PK parameters were evaluated as well. These included age at the time of PK collection, prehydroxyurea laboratory values, weight-for-height z score or body mass index z score, and estimated glomerular filtration rate (eGFR). Weight-for-height z score were determined via the World Health Organization growth charts based on the sex and age of participants through the age of 5 years, and body mass index z score was used for those aged 5 to 19 years.[39] These are collectively referred to as body composition z score. eGFR was calculated based on the pediatric bedside Schwartz equation using serum creatinine and height.[40]

Statistical analysis

Descriptive statistics were generated for population characteristics and PK parameters. Continuous variables were summarized as mean and standard deviation (SD) or median and interquartile range, and categorical variables were summarized as count and percentage. Comparisons among regions were conducted via 1-way analysis of variance for continuous variables. Nominal P values are reported for each variable being compared. Because each preplanned comparison presented stands on its own, no adjustments were performed across the set of reported P values.[41-44]

A generalized linear regression model was applied to investigate whether patient characteristics were associated with PK parameters. In multivariable analysis, significant terms were retained based on backward elimination model selection with .2 significance entry and stay level. A 2-sided P value of <.05 was used to determine the significance of variables in all analyses. Data analyses were performed using SAS version 9.4 (SAS Institute, Cary, NC).

Results

Participant characteristics

PK profiles from 451 individual children were analyzed:146 from the United States, 265 from Africa, and 40 from the Caribbean (Table 2). There were 205 females (45.5%), and the median age at sample collection was 6.8 years (interquartile range, 3.8-9.8; range, 0.5-19.5). Due to variation in trial methods and eligibility, children included from the Caribbean were comparatively older (mean ± SD, 9.7 ± 2.6 years [Caribbean] vs 6.7 ± 2.8 years [Africa] and 6.7 ± 5.3 years [United States]; $P < .001$). Participants from the United States were less underweight, with 15.4% having body composition z scores below −1 compared with 22.5% of those from the Caribbean and 34.4% of those from Africa ($P < .001$). The baseline serum creatinine values in children from the United States were lower (0.31 ± 0.10 mg/dL) than in those from Africa (0.40 ± 0.11 mg/dL) and the Caribbean (0.44 ± 0.07 mg/dL) ($P < .001$). eGFR values were higher in participants from the United States (161 ± 47 mL/min per 1.73 m^2) than in those from Africa (113.6 ± 38.4 mL/min per 1.73 m^2) and the Caribbean (116.4 ± 17.0 mL/min per 1.73 m^2) ($P < .001$).

PK parameters

Due to variation in trial design and methods, the hydroxyurea dose at the time of PK collection in the United States differed from that in the other regions (20.1 ± 1.5 mg/kg [United States]

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Table 2. Participant characteristics, PK-predicted dose, and PK parameters by global region

	Overall (N = 451)	United States (n = 146)	Africa (n = 265)	Caribbean (n = 40)	*P* value
Participant variables					
Female, n (%)	204 (45.2)	57 (39.0)	130 (49.1)	17 (42.5)	.14
Age, y	7.0 (3.9)	6.7 (5.4)	6.7 (2.8)	9.7 (2.6)	<.001
Body composition z score	−0.36 (1.04)	−0.05 (1.16)	−0.50 (0.99)	−0.52 (0.79)	<.001
Body composition z score of less than −1, n (%)	118 (26.8)	21 (15.44)	88 (34.4)	9 (22.50)	<.001
Creatinine, mg/dL	0.37 (0.12)	0.31 (0.10)	0.40 (0.11)	0.44 (0.07)	<.001
eGFR, mL/min	127.8 (40.7)	161.5 (47.1)	112.8 (28.0)	116.4 (17.0)	<.001
Laboratory parameters					
Hb, g/dL	7.9 (1.2)	8.4 (1.3)	7.5 (1.0)	8.2 (1.2)	<.001
HbF, %	13.5 (8.8)	15.7 (11.8)	12.4 (6.7)	13.3 (7.0)	.348
ARC, $\times 10^9$/L	341 (148)	281 (116)	385 (155)	270 (91)	<.001
Platelets, $\times 10^9$/L	401 (164)	399 (183)	398 (159)	422 (118)	.688
ANC, $\times 10^9$/L	6.1 (2.9)	5.4 (3.1)	6.6 (2.9)	5.1 (2.2)	<.001
PK parameters					
Administered dose, mg/kg	22.5 (5.2)	20.1 (1.5)	23.6 (5.6)	24.3 (7.1)	<.001
AUC, mg•h/L	97.9 (39.8)	84.5 (22.3)	102.7 (44.5)	114.9 (44.2)	<.001
Ratio of test dose vs AUC	4.3 (1.4)	4.2 (1.1)	4.3 (1.5)	4.7 (1.3)	.107
Recommended dose, mg/kg per d	27.1 (6.2)	26.6 (5.9)	27.6 (6.5)	25.2 (4.7)	.038
Absorption, h^{-1}	7.7 (8.4)	8.7 (6.5)	7.6 (9.5)	5.4 (6.2)	.072
Volume of distribution, L/70 kg	48.2 (9.4)	50.5 (8.4)	46.8 (9.9)	49.4 (7.7)	.001
Clearance, L/h per 70 kg	20.3 (5.7)	19.6 (4.0)	20.7 (6.4)	20.2 (5.5)	.170

All values are presented as mean (SD) unless otherwise specified.
ANC, absolute neutrophil count.

vs 23.6 ± 5.6 mg/kg [Africa] and 24.4 ± 7.1 mg/kg [Caribbean]; $P < .001$), resulting in expected differences in AUC estimates. For PK testing, the participants from the Caribbean received their individual MTD as determined clinically based on dose escalation (24.3 ± 7.1 mg/kg), and their mean AUC was 114.9 ± 44 mg•h/L, nearly identical to the target MTD exposure of 115 mg•h/L. The participants from Africa were administered their current doses (23.6 ± 5.6 mg/kg), resulting in an AUC of 102.7 ± 44.5 mg•h/L. Although the doses administered for individual PK estimation differed, the ratio of dose administered to the AUC did not differ across regions ($P = .11$).

Regarding other hydroxyurea PK parameters, there was a slightly lower volume of distribution among those from Africa (46.8 ± 9.9 L/70 kg) than among those from the Caribbean (49.4 ± 7.7 L/70 kg) and the United States (50.5 ± 8.4 L/70 kg; $P = .001$; Figure 1A). There were no significant differences in the absorption rate among regions (7.7 ± 8.4 [Africa], 7.6 ± 9.5 [Caribbean], and 5.4 ± 6.2 h^{-1} [United States]; $P = .072$; Figure 1B). Despite differences in serum creatinine and eGFR, there was no difference in hydroxyurea clearance across regions (20.3 ± 5.7 [Africa], 20.7 ± 6.4 [Caribbean], and 20.2 ± 5.5 L/h per 70 kg [United States]; $P = .17$; Figure 1C).

PK-recommended dose

Overall, the mean PK-recommended dose to achieve an AUC of 115 mg•h/L for the entire cohort was 27.1 ± 6.3 mg/kg per day. The PK-recommended dose was normally distributed around the means for all regions (Figure 1D). One-third of participants (33.4%) had a PK-recommended dose between 25 and 30 mg/kg per day. Slightly fewer participants (28.6%) had recommended doses of 20 to 25 mg/kg per day, and 18.8% had recommended doses of 30 to 35 mg/kg per day. Only 9.0% and 9.8% had a recommended dose <20 mg/kg per day or >35 mg/kg per day, respectively, with similar distributions across regions.

The PK-recommended doses to achieve target hydroxyurea MTD exposure were clinically similar across the 3 regions, but the values were statistically different. The PK-recommended dose (mean ± SD) for children from Africa was 27.6 ± 6.5 mg/kg per day, compared to 26.6 ± 5.9 mg/kg per day for those from the United States and 25.2 ± 4.7 mg/kg per day for those from the Caribbean ($P = .038$).

Associations with laboratory and clinical variables

The results of the statistical analyses are shown in Table 3. In univariate analysis, age, creatinine, and absolute neutrophil count were negatively associated with the PK-recommended dose, whereas HbF and absolute reticulocyte count (ARC) were positively associated with the PK-recommended dose. Body composition z score, eGFR, baseline hemoglobin, and baseline platelet count did not influence the recommended dose.

In multivariable analysis with the full model including all independent variables and in the reduced model using backward elimination, age was negatively associated with the PK-recommended

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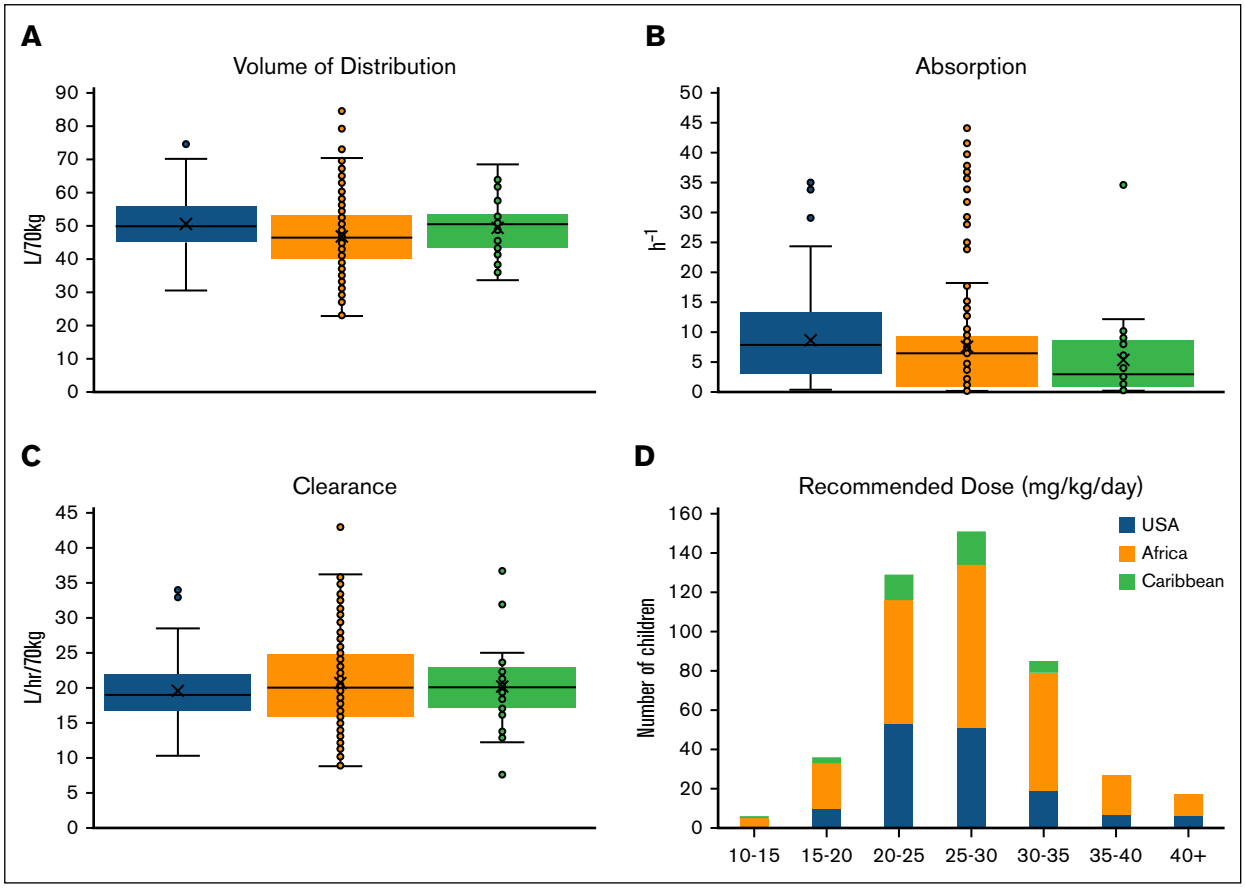

Figure 1. Hydroxyurea PK parameters and predicted PK-guided daily doses. (A) A box-and-whisker plot showing the distribution of volume of distribution among individuals in the United States, Africa, and the Caribbean. (B) A box-and-whisker plot showing the distribution of rates of absorption. (C) A box-and-whisker plot showing the distribution of clearance. (D) A stacked histogram showing the similar distribution of PK-recommended doses (milligram per kilogram per day) across children from the United States, Africa, and the Caribbean.

dose and ARC was positively associated after adjusting for other covariates included in the model. For every 10-year increase in age at the time of PK collection between the age of 0.5 and 19.5 years, the recommended dose decreased by 0.5 mg/kg per day (standard error of the estimate, −0.46 ± 0.07; 95% confidence interval, −0.60 to −0.32; $P < .001$). For every 100 × 10^9/L increase in ARC, the recommended dose increases by 0.9 mg/kg per day (standard error of the estimate, 0.87 ± 0.19; 95% confidence interval, 0.50-1.23; $P < .001$). Associations with creatinine, HbF, and absolute neutrophil count with PK-recommended dose did not remain significant after adjusting for other covariates included in the model.

Discussion

In this large, multinational analysis, we compared the hydroxyurea PK profiles of 451 children with SCA from the United States, Africa, and the Caribbean. In addition to differences in environmental factors and exposures, these patient populations also varied in age, renal function, body composition z score, disease severity, and PK sample collection and analysis methods. Despite these differences, we found remarkable similarities in hydroxyurea PK parameters across global regions.

The average PK-predicted hydroxyurea doses to achieve MTD were 27.6 ± 6.5, 26.6 ± 5.9, and 25.2 ± 4.7 mg/kg per day in the African, US, and Caribbean cohorts, respectively. Although these differences were statistically significant, they were of minimal clinical significance, considering they were within 1 to 2 mg/kg per day, and many treatment protocols allow ±2.5 mg/kg per day or more to accommodate available capsule sizes, of which 500-mg capsules are the most widely available.[10] The PK-recommended doses from this analysis align well with the ultimate MTD achieved through dose escalation for participants in the included trials: HUSTLE, 23.6 mg/kg for "old cohort" and 26.6 mg/kg for "new cohort"; TREAT, 27.6 mg/kg; REACH, 28.2 mg/kg; NOHARM, 29.5 mg/kg; and EXTEND, 25.4 mg/kg.[5,12,13,17,27,37]

In this analysis, the PK-recommended doses were normally distributed around the means in all regions. Over 90% of the children had PK results that predict MTD of >20 mg/kg per day, and over 60% of children were predicted to have MTD of ≥25 mg/kg per day. These results may be used to select optimal starting doses in places where PK testing is unavailable.

The distribution of predicted MTD aligns with recent hydroxyurea pharmacogenetic studies in Angola and Tanzania finding that approximately one-third of patients have inadequate HbF

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Table 3. Summary of univariate and multivariable analyses evaluating the associations between clinical factors and PK-recommended hydroxyurea dose

Predictors	Estimate (SE)	95% CI	*P* value
Univariate analysis*			
Age per 1 year	−0.47 (0.07)	−0.61 to −0.33	**<.001**
Body composition z score	−0.09 (0.28)	−0.65 to 0.46	.738
Creatinine per 1 mg/dL	−11.60 (2.49)	−16.50 to −6.70	**<.001**
eGFR per 100 mL/min per 1.73 m^2	−0.36 (0.73)	−1.80 to 1.07	.619
Hb per 1 g/dL	0.28 (0.24)	−0.20 to 0.76	.247
HbF per 10%	0.13 (0.03)	0.07-0.20	**<.001**
ARC per 100 × 10^9/L	0.89 (0.19)	0.5-1.28	**<.001**
Platelets per 100 × 10^9/L	−0.23 (0.18)	−0.58 to 0.12	.200
ANC per 1 × 10^9/L	−0.14 (0.10)	−0.34 to 0.06	.159
Full multivariable model including all independent variables*			
Age per 1 year	−0.29 (0.11)	−0.50 to −0.08	**.008**
Creatinine per 1 mg/dL	−4.39 (3.08)	−10.45 to 1.67	.155
HbF per 10%	0.07 (0.04)	−0.01 to 0.14	.100
ARC per 100 × 10^9/L	0.96 (0.20)	0.56-1.36	**<.001**
ANC per 1 × 10^9/L	−0.11 (0.10)	−0.32 to 0.09	.265
Reduced multivariable model using backward elimination†			
Age per 1 year	−0.46 (0.07)	−0.60 to −0.32	**<.001**
ARC per 100 × 10^9/L	0.87 (0.19)	0.50-1.23	**<.001**

CI, confidence interval; SE, standard error of the estimate. Bold values are considered significant with $P < 0.05$.
*All independent variables with *P* value <.20 are included in the full multivariable model.
†Significant terms were retained based on backward elimination model selection with .2 significance entry and stay level.

responses to fixed, moderate hydroxyurea dosing (20 mg/kg per day).[45,46] These patients were labeled "poor responders"; however, whether they might have faster hydroxyurea elimination that leads to higher hydroxyurea requirements was not analyzed. The study in Tanzania identified that among 148 individuals with SCA, those with inadequate HbF responses had more pathogenic and rare mutations than those with adequate responses. There were also more variants in genes associated with nitric oxide and the hydroxyurea pathway, including *NOS1* and *ARG2*, and with drug metabolism more generally, including *CYP2E1* and *CYP2C9*.[46] Future studies should pair PK and pharmacogenetic analyses to further elucidate whether "poor responders" are predisposed to low HbF responses (pharmacodynamic differences) or whether they require higher hydroxyurea doses (PK differences).

Children from Africa had lower volumes of distribution than those from the other regions, which may reflect differences in their body composition. Children from Africa and the Caribbean were on average more underweight than those from the United States. Among those from Africa, 34% met the criteria for being undernourished (body composition z score less than −1), in contrast to 23% from the Caribbean and 15% from the United States. However, in regression analysis, there was no impact of body composition z score on the PK-recommended hydroxyurea dose. The impact of splenomegaly on body composition was not analyzed in this study, but it may be an area for future research.

There were no differences noted in hydroxyurea absorption across the global regions or study methods. Prior studies have demonstrated excellent oral bioavailability of hydroxyurea and equivalence between hydroxyurea liquid and capsules as well as between the fed or fasted state.[17,31,32] Studies have also demonstrated variable (fast and slow) absorption profiles based on maximum concentration (C_{max}) and time to C_{max}.[17] Hydroxyurea is a small molecule considered to passively diffuse across membranes, but investigation into the impact of single nucleotide polymorphisms in genes associated with urea transporters (*UTA* and *UTB*) also demonstrates facilitated transport.[17,47] In this analysis, time to C_{max} and C_{max} were not analyzed, but future PK studies may evaluate the distribution of absorption profiles.

There was also no difference in the rate of hydroxyurea clearance across regions despite differences in eGFR and baseline serum creatinine values. In univariate analysis, serum creatinine was negatively correlated with the PK-recommended dose, but neither serum creatinine nor eGFR were significant in multivariable analysis. This lack of correlation between hydroxyurea clearance and eGFR may reflect a limitation in accurately estimating hyperfiltrative renal function and limitations in the of the Schwartz equation in people with SCA, which may be even further complicated by wasting, stunting, or infections.[48,49] Hydroxyurea is eliminated through metabolism to urea and through renal clearance with ~30% to 50% excreted unchanged in the urine.[24,36] Those with severely impaired renal function or end-stage renal disease have lower clearance and should be on a reduced hydroxyurea dose due to the potential for drug accumulation and resulting toxicities.[30] Previous studies have shown that serum creatinine is not a reliable predictor of hydroxyurea clearance in children with SCA, but cystatin C was found to be associated with

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hydroxyurea clearance.[26] Baseline cystatin C from the included trial participants was not available for this analysis, but its inclusion should be considered in future studies.

Increased age at PK collection was associated with a slight decrease in the PK-recommended dose of 0.5 mg/kg per day per 10-year increase in age among infants, children, and adolescents. Although this result is statistically significant, the clinical implication is that hydroxyurea PK change minimally throughout childhood. It should be noted that there was not an even distribution of age groups across this study, with 102 (22%) participants aged >10 years and 349 (77.4%) aged <10 years. Older patients were more often in the HUSTLE or EXTEND trials; therefore, they met trial eligibility based on increased disease severity of burden of SCA complications for HUSTLE and elevated transcranial Doppler velocity for EXTEND. This negative association should not be extrapolated beyond the pediatric and adolescent age range in this study (aged 0.5-19.5 years), because it has not been demonstrated in PK studies of adults.

ARC had a positive association with an increased PK-recommended dose of 0.9 per 100 × 10^9/L, but hemoglobin and HbF did not. This association has been previously identified in the HUSTLE trial in predicting MTD.[50] It is unclear whether this result is due to an impact of hemolysis or robust bone marrow function or whether it is a proxy for erythropoietic drive stimulated by the kidneys. Erythropoietin levels, as well as hemolysis markers, should be evaluated in future PK studies.

There are several limitations to this study. The PK collections were planned prospectively, but the current comparison across regions and trials was designed later. Consequently, there is variation in PK collection methods; despite the method variation, this comparison suggests that the model is robust and not influenced by differing collection methods. An additional limitation is that all participants were enrolled on a clinical trial and may not be comparable to the larger SCA population, but the multinational nature of this analysis and the greater sample size than in previously published PK studies may increase the generalizability of the findings. Additional global populations, such as those in other regions of Africa (Central, Western, and Northern), India, South America, and the Middle East, exposed to differing environmental and genetic factors may have different PK parameters and should be studied in the future.

In clinical trials, it can take up to 12 months to determine an individualized MTD.[27] In clinical practice, logistical barriers, alternate clinical etiologies for cytopenias, and family or clinician discomfort may impede dose escalation and prevent patients from being treated on an optimal dose.[19,20,51] Although low or moderate doses of hydroxyurea have measurable benefits, dose escalation to MTD provides greater protection against SCA complications.[1,7,52,53] Furthermore, some patients and their providers abandon hydroxyurea due to the lack of clinical or laboratory benefit, but it is unclear for how many this lack of perceived benefit is due to suboptimal dosing or inconsistent adherence.[19-21]

PK-guided dosing strategies addresses some of these challenges and shifts the burden of determining an individual's dose from a protracted, stepwise dose escalation scheme to a shorter process that identifies MTD at treatment initiation.[25] Due to the potential benefits of PK-guided dosing with this relatively affordable oral medication, several trials are underway to evaluate the feasibility, safety, and benefits of PK-guided hydroxyurea dosing using HdxSim for people with SCA in Africa, including ADAPT (NCT05662098),[54] SPHERE (NCT03948867),[46] PUSH UP (NCT05285917),[55] and the REACH extension (NCT06171217). This study suggests that the PK model and HdxSim have great potential to be successfully applied to those in Africa and the Caribbean as well. If feasible in low-resource settings, PK-guided dosing may allow more children to be treated with hydroxyurea at individualized, optimal doses from the initiation of treatment, thus maximizing the benefits of this potentially life-saving medication. Even without access to PK testing for individual dose determination, the results from this analysis suggest children with SCA in the United States, Africa, and the Caribbean have similar hydroxyurea PK, and that most are predicted to receive maximum benefit from at least 25 mg/kg per day.

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Authorship

Contribution: A.P.-H and R.E.W. conceptualized this study and wrote the manuscript; R.E.W., T.N.W., G.M., P.O.-O., G.P., L.R.S., M.E.R., A.E.R.-M., R.O.O., C.C.J., P.T.M., C.T.Q., S.E.S., and T.S.L. conducted the clinical trials, including overseeing pharmacokinetic (PK) collection; K.E.M. analyzed the hydroxyurea concentrations; A.A.V. developed the optimal sampling concept with the hydroxyurea population PK model and the area under the time-concentration curve target approach used in this study; N.C.P. designed HdxSim, the analytic software used for PK analysis; and all authors contributed to and reviewed the manuscript, agreed

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with the contents herein, and had access to the primary clinical trial data from the respective clinical trial.

Conflict-of-interest disclosure: A.P.-H. and R.E.W. were formerly on a health equity advisory board for Novo Nordisk, which does not manufacture hydroxyurea. R.E.W. serves as a medical adviser to Nova Laboratories, Theravia, and Merck Pharmaceuticals; and serves on a data safety monitoring board for clinical trials sponsored by Novartis and Vaxart. A.A.V. works at NDA Partners. The remaining authors declare no competing financial interests.

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